

Coefficient of Determination (R^2) Transcript

Hello everyone, welcome to lesson 2 video 1. In this video, we will quantify the coefficients of determination commonly used methods like R^2 . This tells us how well our regression model explains the variability in dependent variable. Hello all, last time we learnt how to find single best fitting line to our data, right? But that leads to a very crucial question, is that line actually meaningful, right? So the most important question or the most crucial question is, is the line that we have fitted is meaningful at all, right? So we are going to look into goodness of our fit or how good our model is in this entire section, okay.

So to define how good our model is, we define a term which is called as R^2 . And R^2 tells us how much amount of variation, okay, how much amount of variation Y is being explained by X . So it is kind of saying the amount or the proportion of total variance in our dependent variable which is being explained by our X variable. So that's what R^2 means and we can actually break it down into simpler terms.

So let's say for example, there could be these three sources of variation that we can get, okay. If you think of the variation can come from the line that you defined, the variation can come from the error which are something you cannot explain, okay. The variation that can come from what you can explain as well, right.

So the total variation is coming from the line, coming from the explained deviation and coming from the unexplained deviations, okay, which you can actually cannot explain in true sense. So in this case, if you think of this as the data, okay. Now if you use a very simple average estimation, which is what your explained deviation is, now the average $\bar{Y}$ line is the dashed line if you think of, right.

And the predicted value of $\hat{Y}_i$ is what you have here. Now this is Y_i minus $\bar{Y}$, that explained deviation, okay. This is the part which is being explained by the model.

Now there is also some part which is not being explained by the model. This one, the difference between Y_i , the true value and the predicted value which is here. This difference is unexplained deviation, right.

I hope this is pretty clear. This line is the average of all the Y values and what you have as unexplained is, unexplained, sorry, what you have as explained is the difference between your predicted value and the average line, this one, wherein I am hovering over and your unexplained deviation is the difference between the true data point, this is the true data point, the black data point that you have and the corresponding prediction on the line, so that is unexplained deviation. This is also the residual term that we have defined in the last session, right, the error term or the residual term that we have defined in the last session.

The idea is to minimise this unexplained deviation. You can use more sophisticated model or more sophisticated model to minimise this, but you generally, if you will get

to all the data points, you generally will not be able to make the entire error zero. Hence, whatever model that you choose, you will never be 100% accurate.

Now from deviations to sum of deviations. Now we, herein we were discussing about only one deviation, right. Now what if we have all the data points, so we will have to measure the difference of all the deviations, okay, or the sum of all the deviations.

On the left-hand side, you have total sum of squares, okay, which is how far the values are from the average value or how far the y values or the dependent variables values are from its average. And then we squared it up because I told you that some of them could be positive, negative, and hence the summation could nullify them, right. This is, this can be broken down into two terms.

One, sum of squares due to regression line. That is what you saw in the previous thing, right. So what component you are able to explain through the model, that's what is this.

And the component that you are not able to explain through the model, which is sum of squares of error. So the total sum of squares is decomposed into two things, sum of squares due to the model or which is being the total variation. You can also think of it as total variation in the y variable is composed of variation because of what you have fitted as the model and the variation which is coming from the error term, right.

And that's what is also explained here if you think of it in this way, okay. So the total sum of squares or total variance in our data is composed of the variance coming from the model and the variance coming from the error. Because of we have defined the variance coming from the model, okay, we can define total variance in the y variable as R square, which is what I told you in the very first in the opening slide, right.

R square is amount of variation in y variable or independent variable, which is being explained by our input data or our model. At top, we have the amount of variation being explained by the model or our input data or our x variables, which is SSR. At the bottom, we have total sum of squares.

Now, SSR can be rewritten as using the equation here, right. SSR can be rewritten as SST minus SSE, right. If you rearrange the term, hence the entire equation becomes R square is equal to 1 minus sum of squares of errors divided by sum of squares of total sum of squares, right.

So, this becomes your entire formula for R square, which is the amount which tells us about how much amount of variation is being explained by input data. This if you think of is dependent on what is not being explained by the data, which is all fine, right. 1 minus not being explained by the data.

So, this is a simple equation that we use to justify or to represent R square and then when we get a data set to build linear regression, we kind of try to explain what is the value of R square. If R square value is closer to 1, it means that the model is good. If

the R square value is closer to 0, it means that your data is not able to explain the variation of Y very well.

That is why R square is 0, okay. So, that can happen if you think of R square when the only situation wherein R square would be 1 is to have SSE as 0 or all the data points lie all in the straight line because there is no deviations from the line, right. In those situations, only SSE would be 0. So, if this becomes 0, R square becomes 1. In what situation R square can be 0? The only situation where R square would be 0 is when SSE becomes equals to SST, right.

When the numerator becomes equals to the denominator and when the numerator becomes equals to denominator, the situation is the entire variation of Y variable is because of the error. Input data has nothing to do with it. That is the interpretation I wanted all of you to get, okay.

Now, here we have, we are trying to visualise the how does random noise or random error affects the R square. Now, in this particular case, it is all set as 0.8. Let us increase or let us first decrease the variation. You see that if the noise is less, everything is on the data point, the numerator term $1 - \frac{SSE}{SST}$, the error becomes smaller, the numerator term in this example, right.

So, this becomes smaller and because this becomes smaller, your R square would be closer to 1 and that is what you see here in this case. Now, if you have more and more noise, your R square would become less and this is how we justify this equation, this entire equation. So, as I have already told you, R square ranges between 0 to 1 on a low value of R square which is 0, it means that your input variables is not explaining Y variable.

If R square value is closer to 1 on the right hand side, it means that your input variables are explaining the variation in Y variable really well, okay. That is what it means. In general, any R square greater than 0.7, even 0.6 is all good.

It depends on case to case basis. Let us say we are trying to build a model to explain the salaries of individual based on their experiences. So, and we get obviously with experience you can think of there is a high correlation, it has a very high correlation with salary and a very experienced individual can have a very high salary but there could be certain outliers as well.

So, in that case, you can think of R square value to be closer to 0.7, 0.8, okay. I hope this entire session gave you an understanding of what exactly happens within linear regression and how do we find out whether which particular model gives us the best approximation of the fit that we use and that we define using a term which is called as R square which measures the variation of Y explained by our input or independent variables X_s , right. So, R square is nothing but total amount of variation in Y variable or dependent variable which is being explained by the model or X variables, okay.



In summary, the coefficient of determination gives us a measure of model fit by showing how much the variation of the dependent variable is explained by our independent variables, okay. In the next video, we will learn how to test significance of regression models.

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